Royal DSM

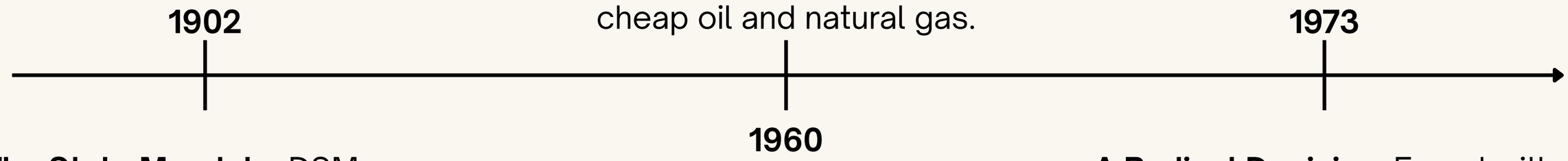
The company that shifted from mining coal to shaping sustainable health and nutrition.

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From National Security to Existential Crisis (1902–1974)

The Rise and Fall of the Coal Paradigm:



The State Mandate: DSM was founded as "Dutch State Mines" to guarantee national energy independence. Its DNA was built on stability and utility-like planning.

The Geological Trap: Deep Dutch coal reserves became too expensive to exploit compared to cheap oil and natural gas.

A Radical Decision: Faced with massive losses, the government decided to close all mines, stripping the company of its original core business.

The Commodity Trap and Cultural Inertia (1975–1990)

Why the First Transformation Failed

- **Fragile New Core**: DSM reinvented itself in petrochemicals and fertilizers, but these products quickly became low-margin commodities.
- **Structural Disadvantage**: High European costs made DSM uncompetitive against global rivals.
- **The "Staatsmijnen" Legacy**: The old state-culture persisted, resulting in slow decision-making, bureaucracy, and risk aversion.
- **The Capability Gap**: The company excelled in bulk chemistry but lacked the specific scientific know-how needed for high-value markets like nutrition.

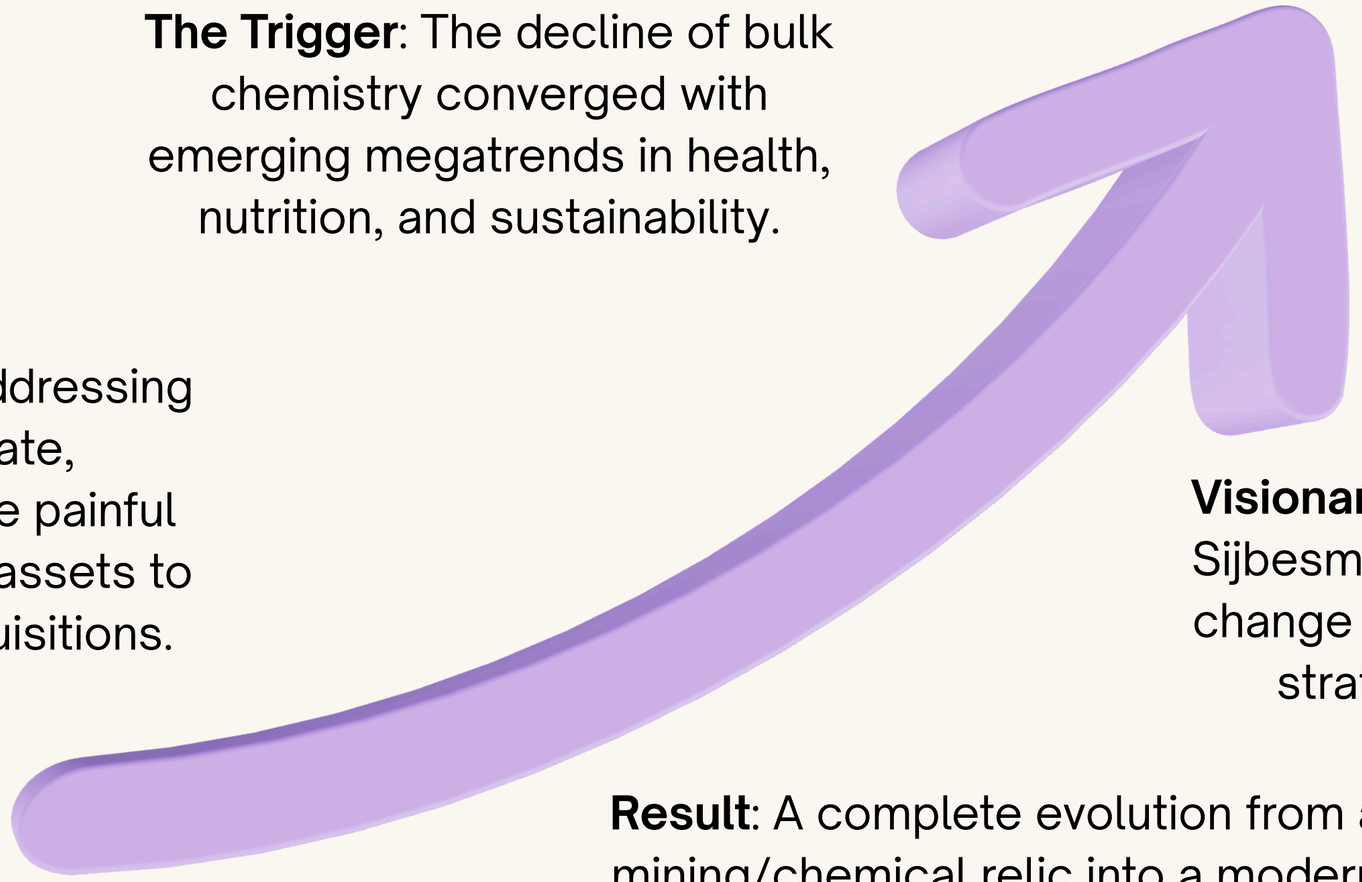


The Strategic Turnaround: Purpose & Megatrends

Leveraging Global Challenges for Business Growth:

The Trigger: The decline of bulk chemistry converged with emerging megatrends in health, nutrition, and sustainability.

Purpose as Strategy: Addressing global issues (climate, malnutrition) justified the painful divestment of chemical assets to fund Life Sciences acquisitions.



Visionary Leadership: CEO Feike Sijbesma (2007–2020) catalyzed change by connecting corporate strategy to social values.

Result: A complete evolution from a mining/chemical relic into a modern Biotechnology leader.

Transformation Goals

STRATEGIC REORIENTATION:
DSM shifted from commodity chemicals to high-value, science-based sectors such as biotechnology, nutritional ingredients, and enzyme-based solutions. This move aimed to reduce dependence on low-margin and cyclical industries.

PURPOSE-DRIVEN IDENTITY:
The company adopted a long-term purpose centered on sustainability, global health, and societal challenges. Initiatives like Brighter Living Solutions reinforced DSM’s commitment to environmental and social value creation.

CAPABILITY BUILDING:
DSM invested significantly in R&D, biotechnology platforms, and scientific talent. Developing new capabilities became essential to support innovation and differentiation in advanced markets.

RESOURCE REALLOCATION:
Financial and human resources were redirected away from declining petrochemical activities toward innovation-intensive businesses. This ensured long-term competitiveness and supported DSM’s transition toward nutrition and bioscience.

Key Phases of DSM's Transformation

1960S–1974

From Coal Mining to Chemicals

DSM exited coal mining (last mine closed in 1974) and shifted to industrial chemicals and fertilizers

1980S–1990S

Entry into Specialty Materials

The company divested low-margin petrochemicals and invested in engineering plastics, resins, and high-performance fibers such as Dyneema.

2010–2023

Purpose-Driven Bioscience Company

DSM embedded sustainability into strategy, exited materials businesses, and merged with Firmenich in 2023.

1990S–2000S

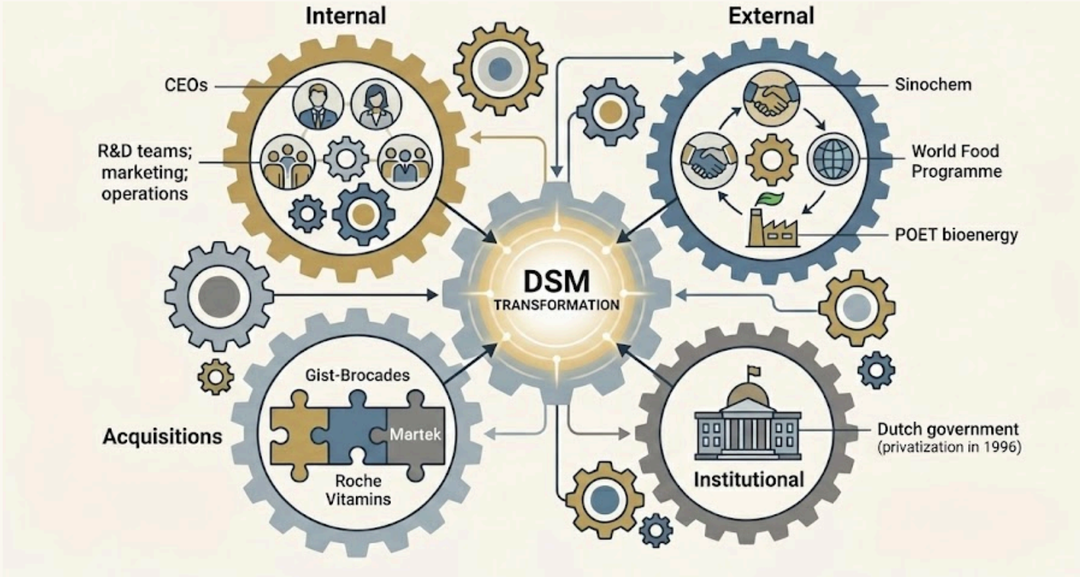
Pivot to Life Sciences and Nutrition

Through major acquisitions (Gist-Brocades, Roche Vitamins, Martek), DSM expanded into biotechnology, enzymes, and nutritional ingredients.



Key Actors in DSM’s Transformation

Internal	CEOs and Board; R&D teams; marketing; operations
External	Sinochem joint venture; World Food Programme; POET bioenergy
Acquisitions	Gist-Brocades, Roche Vitamins, Martek
Institutional	Dutch government (privatization in 1996)



Royal DSM's Transformation Success Factors

1. ***Visionary Leadership & Purpose-Driven Culture***
   - – CEO linked strategy to global challenges like climate and nutrition
   - – "Do Something Meaningful" purpose drove sustainable, impactful innovation

2. ***Innovation & Open Collaboration***
   - – Heavy R&D investment + partnerships with universities and NGOs
   - – Embraced "proudly found elsewhere" to accelerate biotech breakthroughs

3. ***Strong Reputation & Long-Term Investment Appeal***
   - – Top sustainability rankings attracted patient investors
   - – Financial stability amid transformation

Main Barriers to Change

1. **Internal Cultural Resistance**

– Shift from industrial/commodity culture to innovation & sustainability

– Significant retraining, mindset change, and pushback from employees

2. **Pace & Timing Challenges**

- – Balancing slow capability building with fast, disruptive change
- – Risk of organizational disruption vs. missing market opportunities

3. **Short-Term vs Long-Term Financial Tensions**

- – Upfront investments impacted short-term profits
- – Maintaining investor confidence and morale was critical

How Royal DSM Turned Barrier to Opportunity

1. **Low-Margin Anti-Infectives Unit**
   - Anti-infectives business had low profits and was a financial burden
   - DSM partnered with Sinochem to form a joint venture
   - Expanded presence in fast-growing Chinese pharmaceutical market

2. **Declining Coal Mining Operations**
   - Coal reserves depleted, oil & gas became cheaper; mining no longer viable
   - DSM exited coal mining completely by 1974
   - Redirected expertise toward chemicals and biosciences
   - Built new competencies that fueled long-term growth

4. Theoretical Analysis of the Transformation of Royal DSM

Before applying specific frameworks, it is essential to situate DSM’s transformation within the theorized macroeconomic context.

The **BANI** concept offers a powerful interpretative lens for DSM’s decisions.

- **Brittle**: Systems that appear solid can break suddenly. The coal business model became fragile due to the discovery of natural gas and the volatility of oil prices/global competition.

- **Non-linear**: Theory states that in a non-linear context, cause and effect are not proportional. DSM had to abandon the linear logic of commodities and embrace the non-linear logic of bioscience

The REMAIN – REFRAME – REVOLUTION Framework

Remain

For DSM the “REMAIN” asset was not the product (coal or plastics), but the process competence.
The ability to safely manage complex large-scale chemical reactions is a “Remain” asset that allowed DSM to be credible even when entering vitamin production.

Reframing

Chemical catalysis competencies were reframed toward biocatalysis. Chemical engineers were not fired—they were “reconfigured” to manage biological fermenters.

Revolution

The “REVOLUTION” component concerns what is radically new. For DSM, the revolution was not only technological but especially the Business Model.
It shifted from selling volumes (tons of plastics) to selling solutions (health improvement).

Organizational Unlearning: Unlearning to Survive

“Reinvention starts with forgetting” and “Before an organization can change what it does, it must question what it believes.”

Legacy Logics

“Legacy Logics” are like invisible walls that shape decisions.

Forgetting vs Unlearning

Theory stresses the distinction between:

- Forgetting: Passive loss of knowledge (leading to ignorance).
- Unlearning: Active choice that frees “mental bandwidth” for new models.

The Purpose Framework: Beyond the Mission

The third critical framework is the transition toward Corporate Purpose. But first, we have to distinguish mission from purpose:

- Mission: The reason for being as a company (what you do today). “The State Mines” was a mission: extracting coal.

- Purpose: The reason for being for society (a higher objective). “Brighter Living” is a Purpose (Royal DSM’s current motto).



DSM integrated Purpose into corporate statutes and incentive systems, demonstrating how Purpose can act as a “North Star” in a world where markets fluctuate violently.

“Fluid and Continuous” vs “Deep and Bounded” Transformation

There are two visions of transformation:

- Fluid and Continuous
- Deep and Bounded

DSM suggests that successful companies adopt a Hybrid Approach:

- *The major divestments (coal, petrochemicals) were Bounded Transformations (deep, time-limited, top-down).*
- *The evolution toward biosciences was Fluid (continuous, adaptive, cultural).*

Harmonizing these two visions is often the key to success.

Conclusions of the analysis

The analysis of Royal DSM reveals that the company’s success is not accidental but structurally aligned with advanced principles of innovation management.

1

Triple Direction: DSM successfully navigated the convergence of Business, Digital, and Sustainability, proving that sustainability is a growth driver, not a cost.

2

Structural Unlearning: DSM teaches that sometimes Unlearning requires “surgery” (divesting divisions) rather than mere therapy (training).

3

Purpose as Strategy: Royal DSM’s Purpose acted as a resilience mechanism, enabling the company to survive in a BANI world where companies focused solely on short-term profit became “Brittle.”

DSM represents the empirical embodiment of the REMAIN-REFRAME-REVOLUTION model. It managed to Remain true to its scientific roots, Reframe its chemical competencies for a biological world, and Revolutionize its social impact through an authentic Purpose.



Sustainability **01**

DSM used an internal carbon price to make sustainable projects more attractive.

Innovation Ecosystem **02**

DSM innovated faster by partnering externally and buying capabilities

Institutionalizing Agility **03**

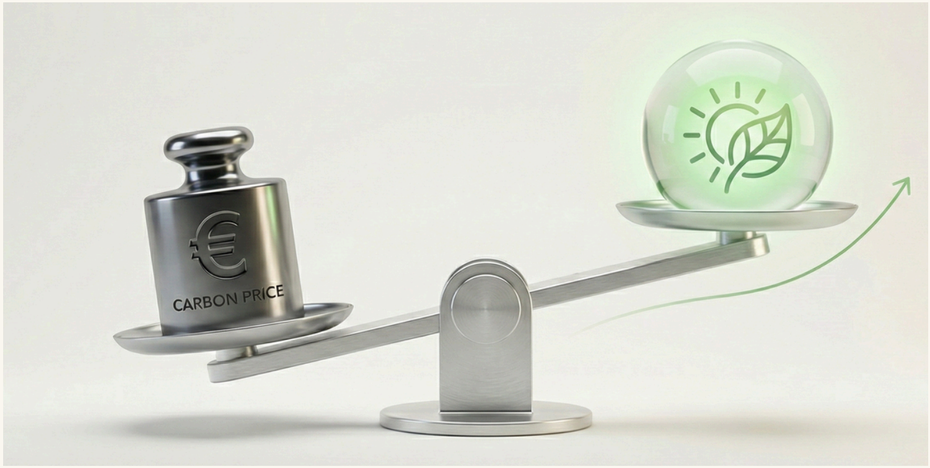
DSM built a culture of constant reinvention by scanning future trends and developing organizational agility.

The Internal Carbon Price as a Strategic Lever

Sustainability is not achieved only across culture, but by changing the internal rules of capital allocation. One of the most technical and interesting implications is the introduction of an **internal carbon price** (set at €50 per ton):

- Every high-emission project started with a "financial handicap" in its budgets, making polluting investments economically disadvantageous

- For managers, this is a lesson in advanced **change management**: It's not enough to ask employees to be sustainable, we need to create internal fiscal mechanisms that make sustainability the only economically rational choice.



The Innovation Ecosystem

The complexity of biosciences makes it impossible for a single company to possess all necessary skills internally:

1. **Open Innovation**: partnering with universities, startups, and even competitors to access skills that are impossible to develop internally alone
2. **Targeted acquisitions**: Acquire companies not to buy market share, but to gain specific capabilities

The modern competitive advantage lies in the ability to manage an external network to reduce risks and development times, not in the exclusive possession of resources



Institutionalizing Agility

The main challenge for managers was not to manage an existing crisis, but to act before the crisis manifested itself:

Horizon Scanning

Managers have to create processes to intercept megatrends (e.g., **climate change**) decades in advance, moving from crisis management to strategic anticipation.

Continuous Reinvention

We must have the courage to abandon consolidated leadership positions (e.g. chemicals) to follow new frontiers before decline, making transformation a **stable** and not episodic capability.

Mobility Culture

Adaptability is trained by encouraging talent rotation and interdisciplinary learning, preventing the organization from becoming stuck in the skills of the past.

What Managers Should Avoid: Common Pitfalls Illustrated by DSM

01

DO NOT HOLD ONTO LEGACY BUSINESSES DUE TO EMOTIONAL OR HISTORICAL ATTACHMENT

LEGACY BUSINESSES CAN BECOME STRUCTURAL OBSTACLES WHEN MARKET LOGICS CHANGE. MANAGERS MUST EVALUATE THEM OBJECTIVELY, NOT SENTIMENTALLY.

02

DO NOT EXPECT CULTURE TO CHANGE SIMPLY BECAUSE THE STRUCTURE CHANGES

ORGANIZATION CHARTS DO NOT SOLVE CULTURAL INERTIA. CULTURE REQUIRES ITS OWN DEDICATED INTERVENTIONS.

03

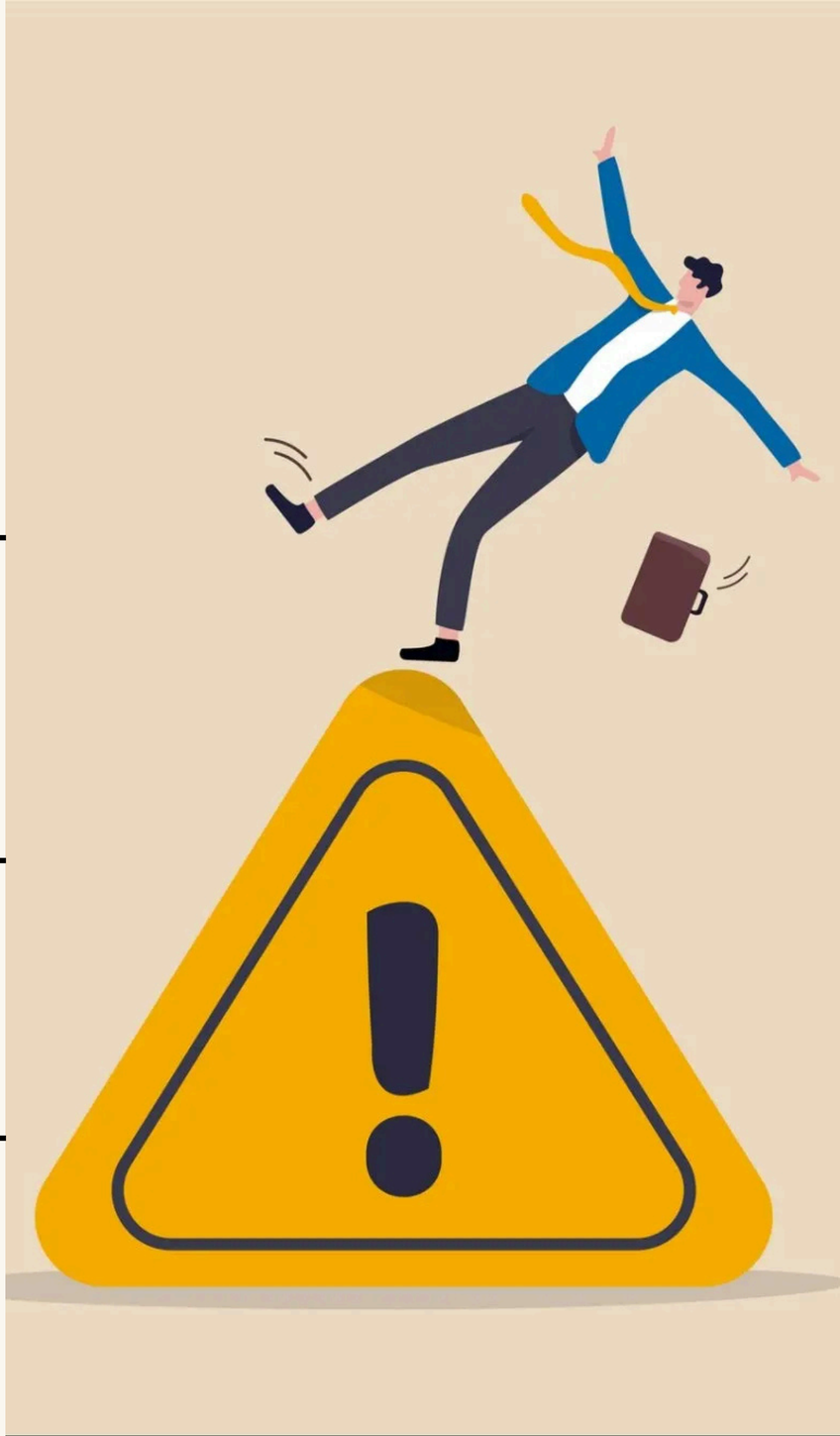
DO NOT TREAT M&A AS AN END IN ITSELF

ACQUISITIONS SHOULD BE CAPABILITY-DRIVEN, NOT SCALE-DRIVEN.

04

DO NOT IGNORE THE TIMING OF TRANSFORMATION

STRATEGIC TIMING IS A COMPETITIVE DIFFERENTIATOR. SLOW RESPONSES CAN MAKE TRANSFORMATION IMPOSSIBLE.



Generalizable Lessons for Organizations Facing Disruption



Purpose Ensures Strategic Consistency in Unpredictable Environments

01



Transformation Should Be Treated as an Ongoing Capability

02



Unlearning Must Precede Learning

03



Capabilities Matter More Than Industries

04



Collaboration Is a More Efficient Route to Innovation Than Isolation

05



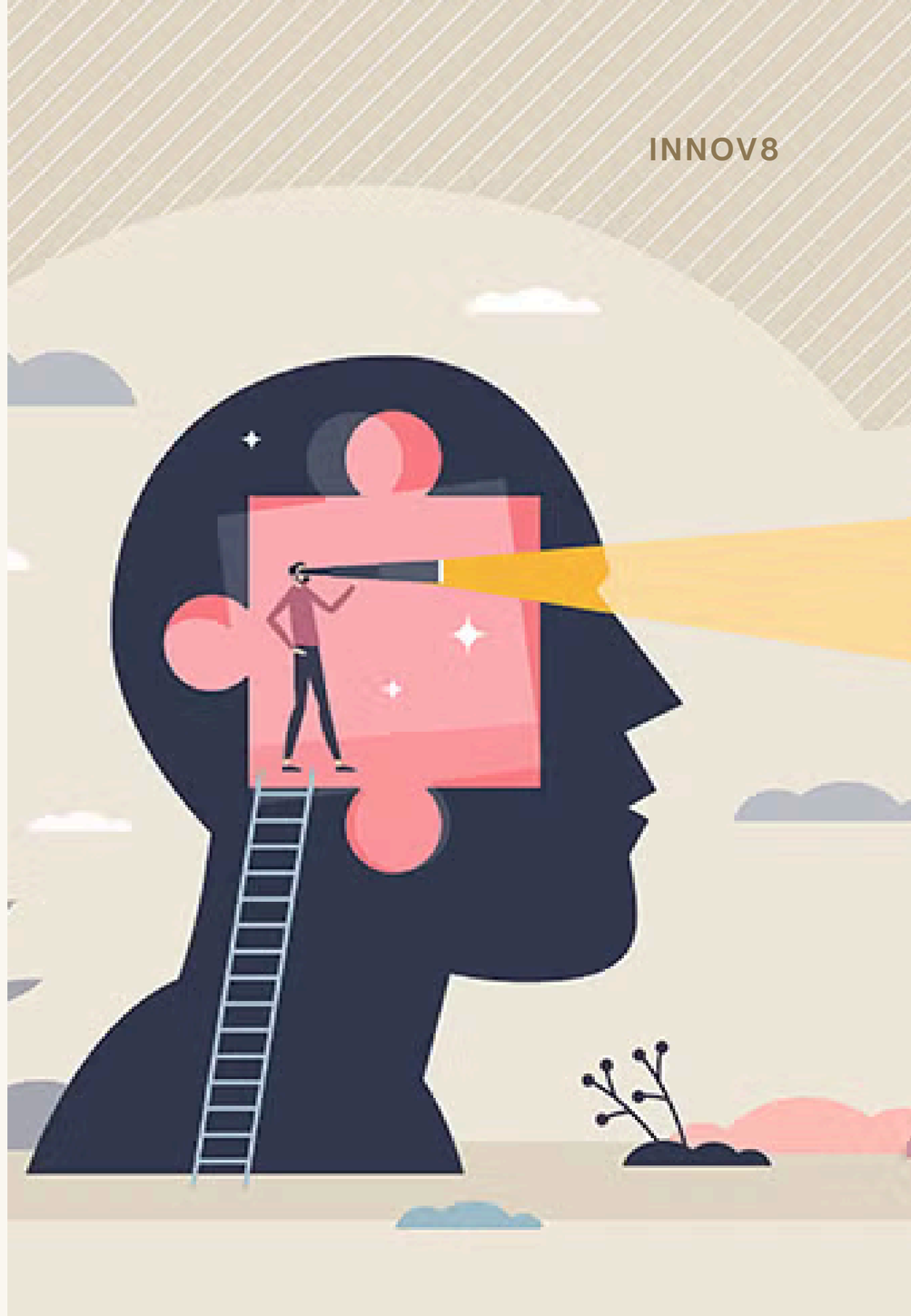
Sustainability Can Be a Competitive Advantage, Not a Cost

06

Final Insight: Transformation Requires Strategic Clarity and Cultural Courage

The central lesson from DSM is that transformation is both structural and psychological. Structural decisions (divestments, acquisitions, capability-building) must be paired with cultural change (unlearning, new narratives, purpose).

Organizations that manage only one dimension tend to fail. DSM succeeded because it combined both.



Thanks for your attention

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